

Bilal Al Rahbi

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PROFESSIONAL SUMMARY

Telecommunications Engineering student at Sultan Qaboos University with strong knowledge in wireless communications, digital signal processing, and intelligent systems. Experienced in developing speaker identification systems using Artificial Neural Networks (ANN) and Gaussian Mixture Models (GMM) with MATLAB-based implementation and real-time GUI development. Hands-on experience in communication infrastructure, troubleshooting, and network systems through hospital technical training. Passionate about telecommunications, AI-driven systems, and real-world engineering applications.

EDUCATION

Bachelor of Engineering in Telecommunications

Sultan Qaboos University —Oman

Graduation: 2026

Relevant Coursework:

Wireless Communication, Digital Signal Processing, Antenna & Propagation, Optical Communication, Telecommunications Security

INTERNSHIP EXPERIENCE

Telecommunications & Electronics Intern

Ibra Hospital - Oman

Duration: 2 Months

- Assisted in maintenance and troubleshooting of UPS systems and electrical protection equipment
 - Supported installation and testing of structured cabling and network infrastructure
 - Worked with hospital communication systems including DECT and nurse call systems
 - Assisted in access control system configuration and maintenance
 - Diagnosed communication and electrical infrastructure faults in operational environments
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FINAL YEAR PROJECT

Speaker Identification System using ANN and GMM

Technologies: MATLAB, ANN, GMM, MFCC, LPCC, PLP, SSC

- Developed a real-time speaker identification system using speech signal processing techniques
- Implemented and compared ANN and GMM classification models
- Designed a MATLAB GUI for training, testing, and live speaker recognition
- Evaluated system performance under multiple datasets and noisy conditions
- Conducted comparative analysis between AI-based and statistical approaches

Key Results:

- Achieved identification accuracy up to 95–100% under different testing conditions
 - Improved system robustness through noise-level (SNR) evaluation and feature optimization
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TECHNICAL SKILLS

Engineering & Telecommunications

Digital Signal Processing, Wireless Communication, Antenna Systems, Embedded Systems, Communication Networks

Software & Tools

MATLAB, Arduino, Proteus

Core Skills

Problem Solving, System Troubleshooting, Technical Analysis, Real-Time System Development

RESEARCH & ACTIVITIES

- Research submission to Student Research Conference 2026 (SRC2026)
- Participation in engineering and technical activities at Sultan Qaboos University